

DAY 12 INTERNSHIP REPORT — MISSING VALUE IDENTIFICATION

Date: 19-09-2026 | **Tools:** Python (Pandas) & Microsoft Excel

1. Executive Overview

For the Day 12 task, missing value identification and data auditing techniques were applied across two distinct datasets. The objective was to evaluate the different approaches required when dealing with programmatically clean data in Python versus real-world messy data in Excel containing both explicit blanks and encoded text markers.

2. Python (Pandas) Analysis: Iris Dataset

Using Pandas, the dataset was loaded via `pd.read_csv()` and audited using `df.isnull().sum()`. The analysis confirmed that the dataset is perfectly clean with zero missing values across all 150 rows.

Column Name	Data Type	Missing Count	Missing %
sepal length (cm)	float64	0	0.0%
sepal width (cm)	float64	0	0.0%
petal length (cm)	float64	0	0.0%
petal width (cm)	float64	0	0.0%
target	object	0	0.0%

3. Excel Analysis: Sales/Orders Dataset

The Excel dataset presented a more complex auditing challenge. Beyond standard empty cells (True Blanks), the dataset contained string-encoded missing markers such as 'Missing', 'N/A', and 'Unknown'. A dedicated Missing-Audit sheet was created to systematically track these discrepancies across the 150-row dataset.

Metric	Count	Notes
True Blanks	45	Standard empty cells natively recognized as blank.
Encoded Markers	14	Cells containing 'Missing', 'N/A', or 'Unknown'.
Total Missing Data	59	Combined data quality issues (39.3% aggregate sum impact).
Most Affected Column	10	Total_Amount column (6.7% missing).

4. Key Takeaways

- 1. Methodology Differences:** Python's `isnull()` is highly efficient for native nulls but requires custom lists to detect encoded text markers. Excel's visual filtering and auditing formulas make string-based anomalies highly visible.
- 2. Data Quality:** The Iris dataset requires zero preprocessing for nulls. In contrast, the Sales dataset requires aggressive cleaning, particularly in financial columns like `Total_Amount` and `Unit_Price`, before statistical analysis can proceed.